

Qatar-1b

R-0356

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170602 · created 2026-07-18T11:33:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457906.89007 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.013
Transit depth (Rp/Rs)^2	3.01 +/- 0.45 [%]
Orbital Inclination (inc)	83.94 +/- 1.51
Airmass coefficient 1 (a1)	1.412 +/- 0.018
Airmass coefficient 2 (a2)	-0.268 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	2.8
Optimal Annulus	8.87
Transit Duration (day)	0.0696 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.013 vs 0.1463 (deviation 1.46σ)
Duration fit vs published	0.0696 d vs 0.0692 d (deviation 0.03σ)
Mid-time O–C	-2.7 min
Depth SNR	9.5
Residual scatter	1.24 %

Light curve

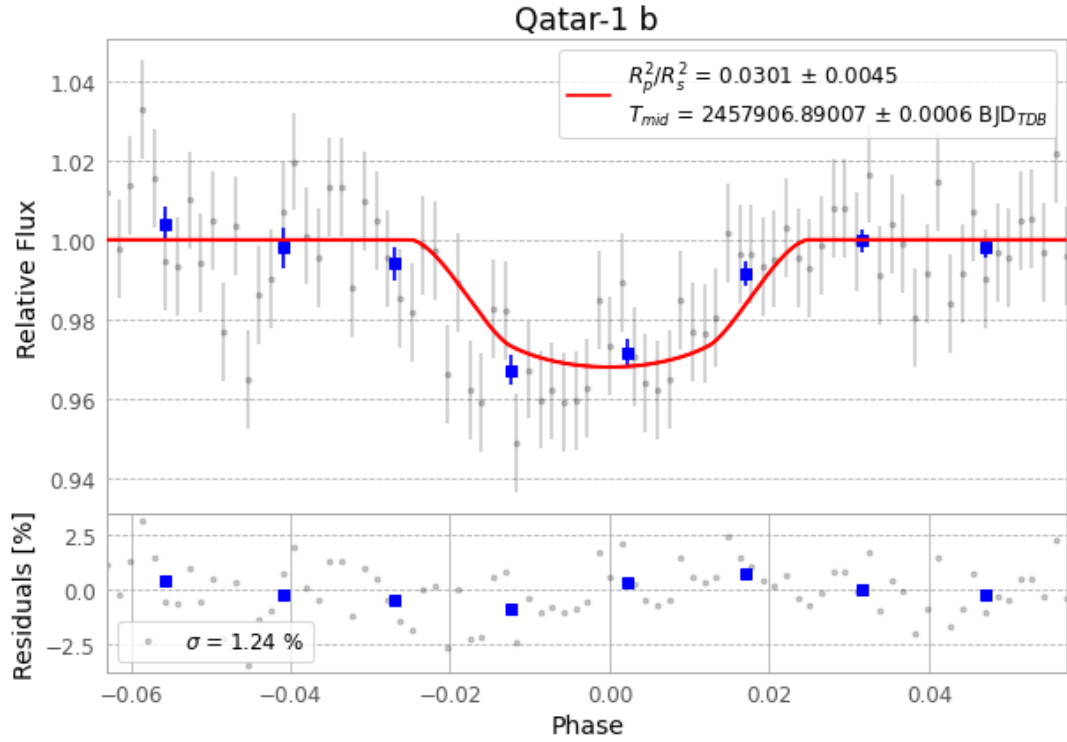


Figure 1 — FinalLightCurve_Qatar-1 b_2017-06-02.png (SHA-256 4e607586be092bfa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [200, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8a1eefd0fce575b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

83 FITS frames (55 MB), Qatar-1170602071212.FITS → Qatar-1170602111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170602.log (083a49d31ca3...), FinalLightCurve_Qatar-1 b_2017-06-02.png (4e607586be09...), FinalLightCurve_Qatar-1 b_2017-06-02.csv (6d73d3bac559...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 11:33Z.